

1 Project: Breast Cancer Segmentation – Semi-Supervised AI System

1.1 TEAM MEMBERS:

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1.2 DATA REPOSITORY (IF APPLICABLE):

The dataset “Advanced-MRI-Breast-Lesions” (<https://doi.org/10.7937/C7X1-YN57>) is from the public database “The Cancer Imaging Archive”. Due to its sheer size of about 600 GB, it is currently available on an external SSD for our project.

1.3 CODE REPOSITORY (IF APPLICABLE):

https://github.com/MelisaAta/AI_Project.git

1.4 PROJECT DESCRIPTION

This project focuses on the development of a semi-supervised artificial intelligence system for automatic tumor region generation in breast MRI images. The project is based on a medical imaging dataset containing more than 620 breast MRI patient cases, where only a small subset of approximately 99 patients includes manually annotated ROI (Region of Interest) tumor masks.

The problem we want to solve is the limited availability of annotated medical imaging data. In medical AI applications, manual segmentation of tumors is a time-consuming and expensive task that requires expert radiologists. As a result, many medical datasets contain only a small number of labeled images, while the majority of the available data remains unlabeled and cannot be directly used for supervised segmentation tasks.

The final goal of this project is to develop a semi-supervised segmentation pipeline capable of learning tumor patterns from the available labeled MRI scans and automatically generating pseudo-ROI masks for unlabeled patient data. The generated masks will then be analyzed and evaluated using segmentation metrics and visual comparison methods. In addition to the AI component, the project also includes data acquisition, preprocessing, exploratory data analysis, visualization, and model evaluation to create a complete advanced data analytics and artificial intelligence workflow.